

RAHUL MITRA

AI ENGINEER PROFILE

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PROFILE SUMMARY

Early-career AI engineer focused on applied model systems, RAG, agentic workflows, NLP, computer vision, and AI-assisted creation pipelines. Built AI products across maritime, healthcare, sustainability, education, and geospatial monitoring.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Artificial Intelligence; 3D Computer Vision; Software Engineering; Simulation; Statistics; Operations Research; Linear Algebra; Probability
- Certificate of Outstanding Performance in 3D Computer Vision (CS4277) - top student in a class of 24, NUS School of Computing.

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories

Jun 2026 - Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.

Data Analytics and Project/Product Management Intern, Abbott Laboratories

Jan 2026 - Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT optimizer for hybrid flow-shop scheduling, translating production objectives and constraints into decision-ready schedules.
- Engineered a non-destructive, 15-stage changeover-data pipeline to normalize operator- and machine-generated anomalies, validating it against five years of previously unseen raw data with zero execution failures.
- Productionized and supported a team-built APC simulator for live manufacturing and engineer training by packaging it in Docker and deploying it to Azure App Service.
- Delivered practical AI upskilling to the regional engineering workforce for day-to-day operational use.

Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Built an end-to-end digital garden-plot balloting platform (admin and public-facing frontends, backend, GIS mapping, and deployment), now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders; awarded the S\$20,000 Sparks Community Innovation Fund.

SELECTED PROJECTS AND COMPETITIONS

Volt Pulse SG - SMU Hack For Cities Finalist

Jan 2026

- Reached Top 8 of 50+ teams with an agentic AI platform for HDB energy-cost tracking using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

Maritime Deficiency Severity Forecasting

2025

- Led model training for a maritime inspection NLP system using a fine-tuned 109M-parameter BERT model, DNN layers, gradient clipping, and ASPIRE 2A supercomputing.

AgeWellLah.AI - NUS HealthHack 2025

2025

- Engineered an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, Flask, OAuth2-secured patient profiles, and PDPA-aware design choices.

SmartExam - GovTech x NTUPC Product Hackathon

2025

- Built a multi-agent autonomous exam generation system with RAG, Next.js, source ingestion, question generation, and education workflow automation.

EthosLens - LifeHack 2025

2025

- Developed backend workflows for an AI research engine that scores brand ethics and sustainability practices using FastAPI, Python, LangChain, and automated evidence retrieval.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Python; BERT; DNN; RAG; AI Agents; GPT-4; Gemini; Vector Search; SEALION; ASPIRE 2A; 3D Computer Vision; Projective & Epipolar Geometry; Absolute Pose; Structure-from-Motion; Bundle Adjustment; Multi-View Stereo; Model Evaluation.

- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.